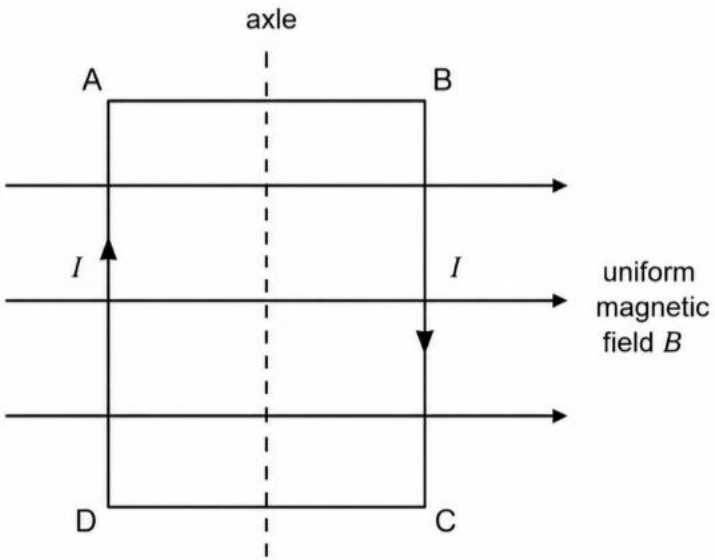
23 A rectangular coil ABCD is mounted on a vertical axle. Its plane is parallel to the field lines of a uniform horizontal magnetic field. Current flows upwards through side DA and downwards through side BC.



When the current is first switched on and before the coil moves, which statement is correct?

A There is no magnetic force on sides AB and CD.
B There is no magnetic force on sides BC and DA.
C Sides BC and DA experience forces in the same direction.
D All four sides experience magnetic forces of equal magnitude.

Ans: A

Use $F = BIL\sin\theta$

Sides AB & CD are parallel to the magnetic field, so the magnetic force is zero.

Sides BC & DA are perpendicular to the magnetic field, but have opposite currents, so the magnetic forces acting on them are in opposite directions.